1 Table T + question q

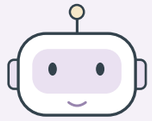
Year	Region	Product	Sales	Cost	Profit
2021	North America	Product A	8,100	5,000	3,100
2021	Europe	Product A	6,200	3,800	2,400
2022	North America	Product A	9,000	5,400	3,600
2022	Europe	Product A	6,700	4,100	2,600
2023	North America	Product A	11,200	4,825	6,375
2023	North America	Product B	8,400	4,900	3,500
…	…	…	…	…	…



What is the total profit of Product A in North America in 2023?

Product A | North America | 2023

2 Predict question type §3.3



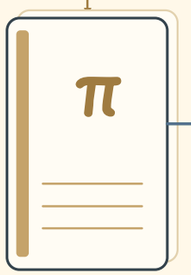
LLM classifier

- Lookup
- Order
- Local Reasoning
- Count
- …

3 Select scope

Fixed policy π · §3.3

Type | Table size



Policy set on validation data

Localized reasoning §3.4

4 Operation-aware retrieval

K = 4 per selector

Infer operations

Filter | Compare | Aggregate | Count | Sort | Lookup

Guide evidence coverage



Relevant columns



All needed rows

5 Aggregate

Support + Compactness

Cols

Rows

Year	Region	Product	Profit
2023	North America	Product A	6,375
2023	North America	Product B	3,500
…	…	…	…

6 Refine · sufficiency



T → T'

✓ Yes: keep -> sub-table

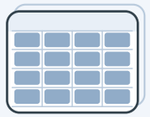
No: expand rows / columns

7 Refined sub-table

Year	Region	Product	Profit
2023	North America	Product A	6,375

(q, T')

Full-table reasoning



q

(q, T)

Bypass

8 Answer generation §3.5



Answer: 6,375